AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION 2010

General Mathematics Paper II

Time allowed: 2 hours 20 minutes Marks 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are ELEVEN questions. Answer TEN questions. Choices are specified inside the paper

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use colour pencils.
   DO NOT use staples, paper clips, glue or correcting fluid.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

Mr. Aslam is planning to invite 90 guests in his daughter's birthday party. A baker told him that a 3 pound cake is good enough for 20 people. According to the baker's estimation, what should be the mass of cake for his guests?

Q.2. (Total 4 Marks)

EITHER

a. Mrs. Rashid owned a gold set of mass 95 gram for last one year. She also has Rs.30,500 as her annual savings. What is the amount of Zakat for her to pay if gold rate is Rs.38,000 per 10 gram? [Note: Rate of Zakat is 2.5% of total annual saving.]

OR

b. Mr. Sajid owns a house and has Rs.9,30,000 in his account. He wants to divide his property and savings among his 4 daughters and 3 sons. He sold his house for Rs.40,00,000. Find the share of each son and daughter, if a son gets twice the share of a daughter.

Q.3. (Total 5 Marks)

EITHER

a. An electrical appliance shop offers 11% discount on a microwave oven, priced Rs.5500. A person bought the oven and paid 2% credit card fee of the discounted price. What is the net saving on this purchase?

OR

b.

i. Ms. Farzana bought 12 sweaters for her shop for Rs.7200. She would like to earn 20% profit for her investment. What will be the selling price of each sweater? Also find her total profit in Rupees. (4 Marks)

ii. 30 is what percent of 50? (1 Mark)

Q.4. (Total 3 Marks)

The cash price of a car is Rs.5, 50,000. Ms. Zareen has purchased the car on lease by making a down payment of Rs.1, 00,000, followed by 24 monthly instalments of Rs.22,000 each.

a. Find the total leasing cost of the car.

b. What is the percentage of additional charges that she has paid on leasing price as compared to actual price?

Q.5. (Total 5 Marks)

The salary schedule for a medical representative of a pharmaceutical company is as follows:

Weekly fixed pay for a representative	Rs.3000
Overtime for one hour	Rs.300
Commission for every order less than Rs.5000	3% of sale
Commission for every order greater than Rs.5000	5% of sale

What will be the salary of a representative, if he has worked for 4 hours extra in a week and has booked the orders of Rs.2000, Rs.11000 and Rs.2500. [Note: commission is calculated for every order separately]

Q.6. (Total 5 Marks)

EITHER

a. If $a = 4\times10^3$, $b = 3.5\times10^4$ and $c = 15{,}000$, then Simplify $\dfrac{a+bc}{b-c}$.

Write your answer in scientific notation.

OR

b. Attempt both part i and ii.

i. By using the following table, find the value of log (37.19). (3 Marks)

LOGARITHM TABLE

	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
35	5441	5453	5465	5478	5490	5502	5514	5527	5539	5551	1	2	4	5	6	7	9	10	11
36	5563	5575	5587	5599	5611	5623	5635	5647	5658	5670	1	2	4	5	6	7	8	10	11
37	5682	5694	5705	5717	5729	5740	5752	5763	5775	5786	1	2	3	5	6	7	8	9	10
38	5798	5809	5821	5832	5843	5855	5866	5877	5888	5899	1	2	3	5	6	7	8	9	10

ii. By using the following table, find the value of antilog (0.31). (2 Marks)

ANTI LOGARITHM TABLE

	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
.30	1995	2000	2004	2009	2014	2018	2023	2028	2032	2037	0	1	1	2	2	3	3	4	4
.31	2042	2046	2051	2056	2061	2065	2070	2075	2080	2084	0	1	1	2	2	3	3	4	4
.32	2089	2094	2099	2104	2109	2113	2118	2123	2128	2133	0	1	1	2	2	3	3	4	4
.33	2138	2143	2148	2153	2158	2163	2168	2173	2178	2183	0	1	1	2	2	3	3	4	4

Q.7. (Total 4 Marks)

The nth term of arithmetic sequence $\frac{3}{2}, \frac{7}{4}, 2, \frac{9}{4}, \ldots\ldots$ is 4. Find *n.*

Q.8. (Total 5 Marks)

EITHER

a. For all three Venn diagrams given below, shade the region $(A \cup B)$ as  and $(A \cup B) \cap C$ as  .

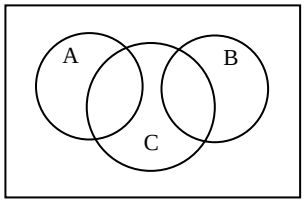


Figure i

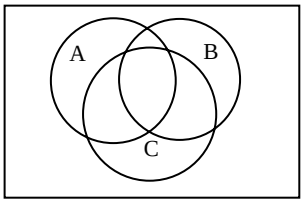


Figure ii

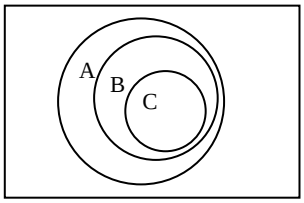


Figure iii

OR

b. Attempt both part i and ii.

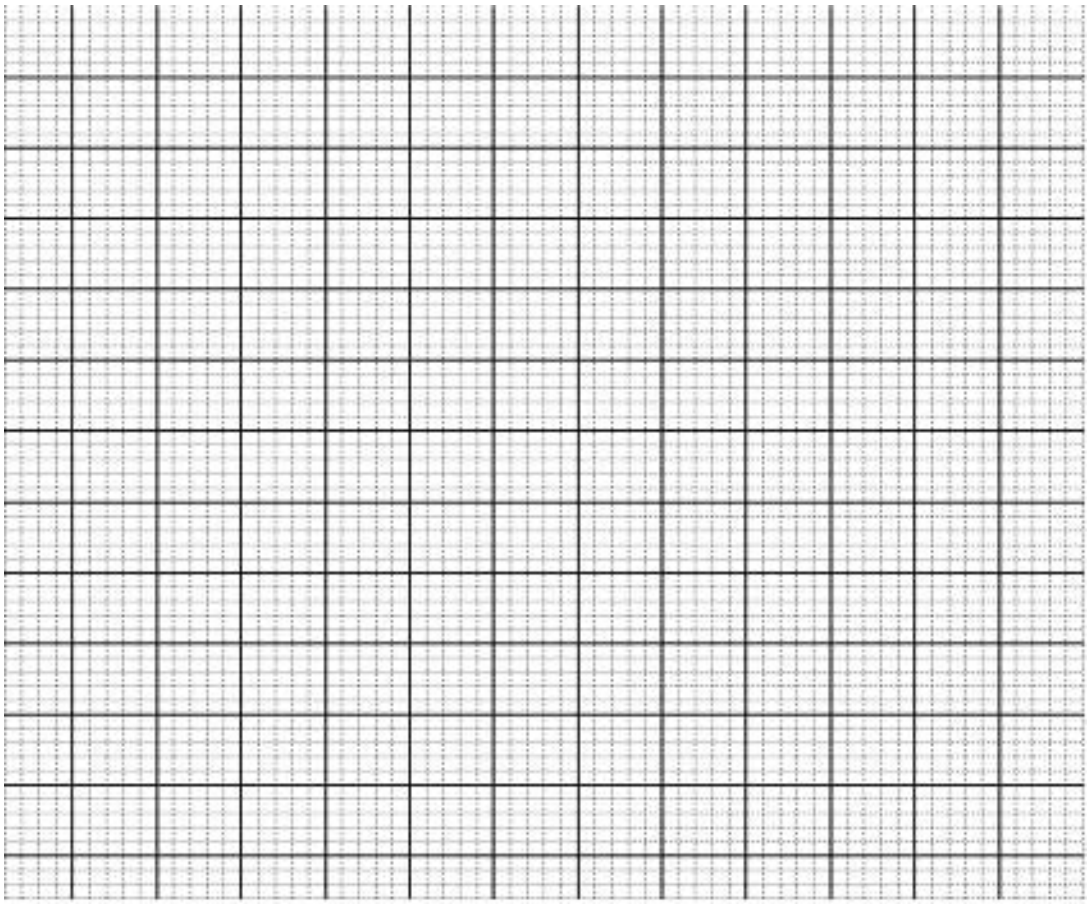
i. If A = {whole number divisible by 3}, then write the set builder form and the tabular form of set A. (2 Marks)

ii. If X = {0,1,3,5,7} and Y = {0,2,8,10,12}, then find $(X \cup Y) - (X \cap Y)$. (3 Marks)

Q.9. (Total 5 Marks)

Complete the following table for pairs of values satisfying the linear equation $3x - 2y = 5$ and hence draw the graph of equation $3x - 2y = 5$ on the following graph.

x	1		
y	–1		

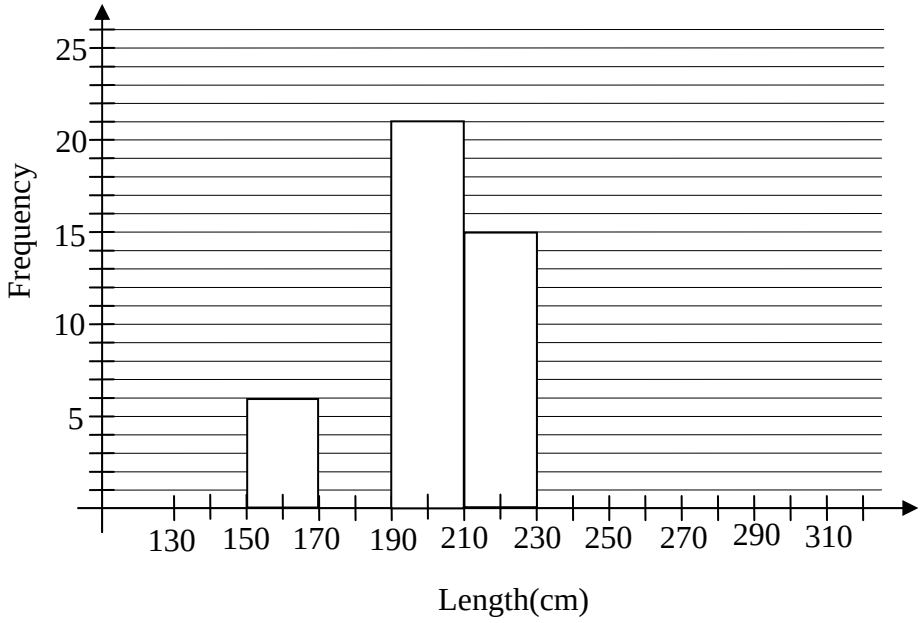


(**ATTEMPT EITHER Q.10 OR Q.11)**

Q.10. (Total 6 Marks)

The table and histogram below are both incomplete. They represent the information about the lengths of sugarcanes after cutting the crop. Complete the table and histogram both.

Lengths (cm)	150-170	170-190	190-210	210-230	230-250
Frequency		10	21		6



a. Find the total number of sugarcanes taken as the sample.

b. How many sugarcanes have lengths between 170 cm and 230 cm.

(**ATTEMPT EITHER Q.10 OR Q.11)**

Q.11. (Total 6 Marks)

Mr. Nadeem is an incharge of internet club of a school. He has recorded the time (in minutes) spent on the internet by each of 12 students in one day. Below are the results:

68	79	91	77	88	82
76	65	82	68	77	86

Find the range, standard deviation and variance of the data.

Please use this page for rough work

Please use this page for rough work

Please use this page for rough work